



Synthesis of three-dimensional nitrogen doped meso/macroporous carbon beads for heterogeneous catalytic solvent-free oxidation of ethylbenzene

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ABSTRACT

A series of nitrogen doped 3D hierarchically porous carbon beads with size of 0.6–0.9 mm are successfully prepared by one-step pyrolysis of dicyandiamide and macroporous resin under different ratio. Dosage of dicyandiamide can effectively control shrinkage of resins during pyrolysis, and favor to create 3D interconnected pore channels and abundant nitrogen doped sites as a self-sacrificing dopant and activating-like agent. The beads display excellent conversion for ethylbenzene to acetophenone as opposed to only calcinating pristine IRA resins (83.5% vs 12.1%), when the ratio reach 1:1. At initial reaction period, linear relationships between concentration of reactants and products are established to evaluate whether the beads are provided with diffusion limitation in mentioned reaction. Structural characterization and reaction kinetic studies indicate that the larger mesoporous pore volume and more meso/macro-pore structure of the beads can expose more accessible active sites and lower diffusion limitation, significantly affecting the initial rates, thereby improving the catalytic performance. In addition, macro-beads in size of millimeter scale can effectively exert the advantages of easy separation and recovery. A new strategy is provided which can keep the diversity of pore scale in porous carbonous precursors during pyrolysis, thereby more effectively applying in heterogeneous catalysis.

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1. Introduction

Aromatic ketones (e.g., acetophenone) are important industrial intermediates for preparation of pharmaceuticals and chemical-products [1–6]. Aromatic ketones have been synthesized by the oxidation of benzylic C–H via homogeneous iron coordination compounds or heterogeneous metal-based materials as catalysts, generally leading to water pollution [3,7]. Therefore, it is meaningful for finding an environmentally friendly, easy to be separated and recyclable catalyst for ethylbenzene (EB) to acetophenone (AcPO).

The nonmetal atoms, especially electron-rich nitrogen atoms,

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can be easy to dope into skeleton of carbonous materials and modifying their properties, thereby extending the potential applications [4,5,8–17]. For structural nitrogen such as pyridinic N (N1) and pyrrolic N (N2), the oxidation of benzyl hydrocarbon bonds shows very low activity, as does the activity of chemical nitrogen (amine or nitrosyl group etc.) [9,18]. Gao and co-workers demonstrate that graphitic N (N3) can cause deformation of ortho-carbon which is easier to activate substrate molecules, further accelerating the EB oxidation reaction process [8]. However, there are few reports on the synthesis of heterogeneous nitrogen-doped three-dimensional hierarchically porous activated carbon (3D-HPC) catalysts which are applied for mentioned liquid reaction. To our best of knowledge, 3D-HPC materials are largely composed of defective graphene-like nanosheets [4,19–21]. Because of their highly stacked together via strong π - π interactions, the stocking of 2D nano-sized sp^2 -dominated building blocks would reduce their porosity and the number of exposed active sites, as well as improve the diffusion resistance of reactants [22–28]. 3D hierarchical